KAIZEN MATHS

Lesson starter

Discrete Random Variables and Probability Distributions

Fundamentals

Name _____

S1

x	0	1	2	3	4
P(X=x)	0.28	0.06	0.17	0.11	0.38

Find P(X = 2)

S2

x	1	2	3	4	5	6
P(X=x)	0.17	0.12	a	0.20	0.17	0.17

Find a

S3

Is this a valid probability distribution?

x	0	1	2	3
P(X=x)	0.33	0.33	0.08	0.26

S4

x	2	3	4	5	6	7
P(X=x)	0.21	0.23	0.18	0.15	0.13	0.10

Find P(X > 5)

Show your working

KAIZEN MATHS

Lesson starter

Discrete Random Variables and Probability Distributions

Fundamentals

Name _____

S1

x	0	1	2	3	4
P(X=x)	0.28	0.06	0.17	0.11	0.38

Find P(X = 2)

S2

x	1	2	3	4	5	6
P(X=x)	0.17	0.12	a	0.20	0.17	0.17

Find a

S3

Is this a valid probability distribution?

x	0	1	2	3
P(X=x)	0.33	0.33	0.08	0.26

S4

x	2	3	4	5	6	7
P(X=x)	0.21	0.23	0.18	0.15	0.13	0.10

Find P(X > 5)

STA-2615-L1 · S

Show your working

KAIZEN MATHS

Exit ticket

Discrete Random Variables and Probability Distributions

Fundamentals

Name _____

EA1

x	1	2	3	4	5
P(X=x)	0.23	0.14	q	0.14	0.44

Find q

EA2

Is this a valid probability distribution?

x	0	1	2
P(X=x)	0.29	0.57	0.29

KAIZEN MATHS

Exit ticket

Discrete Random Variables and Probability Distributions

Fundamentals

Name _____

EB1

x	2	3	4	5
P(X=x)	0.31	0.14	0.28	0.27

Find P(X = 2)

KAIZEN MATHS

Exit ticket

Discrete Random Variables and Probability Distributions

Fundamentals

Name _____

EA1

x	1	2	3	4	5
P(X=x)	0.23	0.14	q	0.14	0.44

Find q

EA2

Is this a valid probability distribution?

x	0	1	2
P(X=x)	0.29	0.57	0.29

STA-2615-L1 · EA

Show your working

KAIZEN MATHS

Exit ticket

Discrete Random Variables and Probability Distributions

Fundamentals

Name _____

EB1

x	2	3	4	5
P(X=x)	0.31	0.14	0.28	0.27

Find P(X = 2)

STA-2615-L1 · EB

Show your working

Teacher answers & working

Discrete Random Variables and Probability Distributions · Fundamentals

STA-2615-L1 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S1

x	0	1	2	3	4
P(X=x)	0.28	0.06	0.17	0.11	0.38

Find P(X = 2)

Answer:

0.17

Working:

Identify the values included

Included values: X = 2

Add the matching probabilities

$P(X=2) = 0.17$

0.17

S2

x	1	2	3	4	5	6
P(X=x)	0.17	0.12	a	0.20	0.17	0.17

Find a

Answer:

$a = 0.17$

Working:

Use the total probability rule

All probabilities must sum to 1.

Add known probabilities

Known sum = 0.83

Subtract from 1

$a = 1 - 0.83 = 0.17$

$a = 0.17$

Teacher answers & working

Discrete Random Variables and Probability Distributions · Fundamentals

STA-2615-L1 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S3

Is this a valid probability distribution?

x	0	1	2	3
P(X=x)	0.33	0.33	0.08	0.26

Answer:

Yes, valid

Working:

Check each probability

Each probability is between 0 and 1.

Add all probabilities

$0.33 + 0.33 + 0.08 + 0.26 = 1.00$

Sum is 1.00, so valid.

S4

x	2	3	4	5	6	7
P(X=x)	0.21	0.23	0.18	0.15	0.13	0.10

Find P(X > 5)

Answer:

0.23

Working:

Identify the values included

Included values: X = 6, X = 7

Add the matching probabilities

$P(X=6) + P(X=7) = 0.13 + 0.10$

0.23

Teacher answers & working

Discrete Random Variables and Probability Distributions · Fundamentals

STA-2615-L1 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

EA1

x	1	2	3	4	5
P(X=x)	0.23	0.14	q	0.14	0.44

Find q

Answer:

$q = 0.05$

Working:

Use the total probability rule

All probabilities must sum to 1.

Add known probabilities

Known sum = 0.95

Subtract from 1

$q = 1 - 0.95 = 0.05$

$q = 0.05$

EA2

Is this a valid probability distribution?

x	0	1	2
P(X=x)	0.29	0.57	0.29

Answer:

No, invalid

Working:

Check each probability

Each probability is between 0 and 1.

Add all probabilities

$0.29 + 0.57 + 0.29 = 1.15$

Sum is not 1.00, so invalid.

Teacher answers & working

Discrete Random Variables and Probability Distributions · Fundamentals

STA-2615-L1 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

EB1

x	2	3	4	5
P(X=x)	0.31	0.14	0.28	0.27

Find P(X = 2)

Answer:

0.31

Working:

Identify the values included

Included values: X = 2

Add the matching probabilities

$P(X=2) = 0.31$

0.31